

A Hybrid Intrusion Detection System using Explainable AI for Enhanced Accuracy and Transparency

Publisher: IEEE [Cite This](#)  PDF

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Abstract	Abstract: Intrusion detection system is highly effective and easy to understand in the situation of ubiquitous cyber threats. The trustworthiness and interpretability of traditional intrusion detection systems are balanced due to their dependence on transparent deep learning (DL) and Machine Learning (ML) methods. To overcome these issues, this research proposes a hybrid IDS which makes use of Explainable AI(XAI). An integration of ML methods like random forest and support vector machine with DL methods like CNN ensure efficient detection accuracy in the system. The decision-making process can be better efficient with the help of XAI methods like SHAP and LIME, which increases transparency. The CICIDS 2017 dataset is used to validate the feature extraction of framework, model training and interpretability components. The result show that proposed method performs better with 98.5% accuracy rate and better precision, recall and F1-score when compared to existing methods. This method provides useful data for securing networks while reducing the number of false positives and negatives. Through the integration of advanced machine learning/deep learning methods with XAI, the proposed IDS ensures dependability, openness and practicality.
Document Sections	
I. INTRODUCTION	
II. LITERATURE SURVEY	
III. PROPOSED METHODOLOGY	
IV. EXPERIMENTAL ANALYSIS	
V. Conclusion	

Authors	Published in: 2025 International Conference on Electronics and Renewable Systems (ICEARS)
Figures	Date of Conference: 11-13 February 2025 DOI: 10.1109/ICEARS64219.2025.10940840
References	Date Added to IEEE Xplore: 02 April 2025 Publisher: IEEE
Keywords	► ISBN Information: Conference Location: Tuticorin, India
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